1) #include <stdio.h>

```
#define PI 3.14

void calculate(int a, int b);

void calculate(int a, int b)
{
int sum = a + b;

#ifdef DEBUG
printf("Debug: a = %d, b = %d\n", a, b);
printf("Debug: Sum = %d\n", sum);
#endif

printf("Sum = %d\n", sum);
}

int main()
{
calculate(10, 20);
return 0;
}
```

OUTPUT :

```
Sum = 30
```

2) #include <stdio.h>

```
int main()
{
int age, amount;
long int population;
float marks;
char grade;

printf("Enter age: ");
if (scanf("%d", &age) != 1)
{
printf("Invalid input!\n");
return 1;
}

printf("Enter population: ");
if (scanf("%ld", &population) != 1)
{
printf("Invalid input!\n");
return 1;
}

printf("Enter marks: ");
if (scanf("%f", &marks) != 1)
```

```
{
printf("Invalid input!\n");
return 1;
}

printf("Enter grade: ");
scanf(" %c", &grade);

printf("Enter amount in paise: ");
if (scanf("%d", &amount) != 1)
{
printf("Invalid input!\n");
return 1;
}

printf("\n--- Student Details ---\n");
printf("Age = %d\n", age);
printf("Population = %ld\n", population);
printf("Marks = %.2f\n", marks);
printf("Grade = %c\n", grade);
printf("Amount = Rs %d.%02d\n",
amount / 100, amount % 100);

return 0;
}
```

```
OUTPUT :
Enter age: 20
Enter population: 50000
Enter marks: 85.5
Enter grade: A
Enter amount in paise: 16250

--- Student Details ---
Age = 20
Population = 50000
Marks = 85.50
Grade = A
Amount = Rs 162.50
```

```
3)#include <stdio.h>

int main()
{
int a = 5, b = 2;
float x = 5.0, y = 2.0;
long int num = 5000000000L;
float charge = 12.567;
```

```
printf("Integer division = %d\n", a / b);

printf("Float division = %.2f\n", (float)a / b);

printf("Type conversion = %.2f\n", a + y);

printf("Casting long to int = %d\n", (int)num);

printf("Charge without rounding = %.3f\n", charge);

printf("Charge rounded to 2 decimals = %.2f\n", charge);

return 0;
}
```

```
OUTPUT :
Integer division = 2
Float division = 2.50
Type conversion = 7.00
Casting long to int = 705032704
Charge without rounding = 12.567
Charge rounded to 2 decimals = 12.57
```

4)

```
#include <stdio.h>

#define VALUE 10

int main()
{
int a = VALUE;
int b = 20;

printf("Sum = %d\n", a + b);

return 0;
}
```

```
OUTPUT :
Sum = 30
```